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PROJECT: Adversarial Robustness in Semiconductor AOI
Systems
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[PROBLEM STATEMENT]
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Automated Optical Inspection (AOI) systems in chip fabrication are

vulnerable to mathematically calculated 'Adversarial Noise'. This project

demonstrates how a 90% accurate system can be sabotaged to <20% accuracy

using the Fast Gradient Sign Method (FGSM), hiding critical hardware flaws.

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[METHODOLOGY]
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- Dataset: Fashion-MNIST (used as a Structural Proxy for wafer patterns).

- Attack: FGSM (Fast Gradient Sign Method) at Epsilon = 0.05.

- Metrics: Accuracy, Precision (1-FPR), Recall (TPR), F1-Score, and ROC/AUC.

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/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
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```
    super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

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[Step 1] Training Inspection System...
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```
32/32 _____ 0s 9ms/step
```

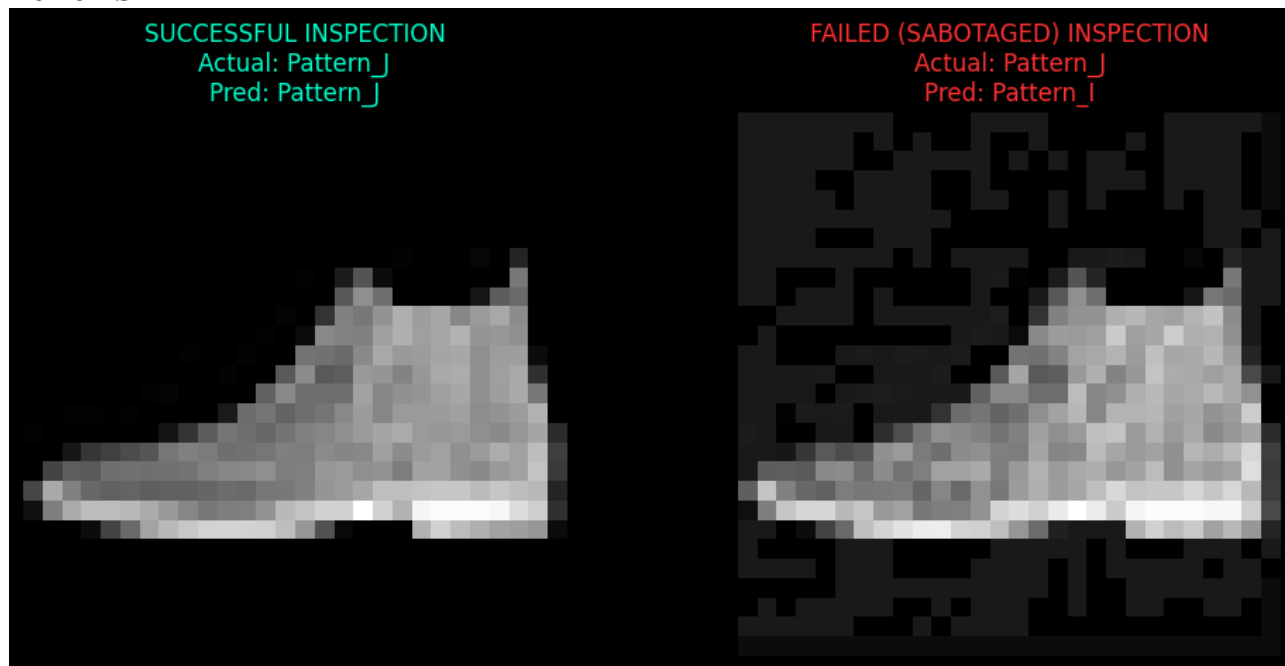
```
32/32 _____ 0s 2ms/step
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SECTION 1: VISUAL EVIDENCE OF SYSTEM BREACH
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DESCRIPTION: This comparison shows how a standard 'Pattern' (the test shoe) is correctly identified under normal conditions but misclassified after applying FGSM noise. Notice the noise is nearly invisible to humans.



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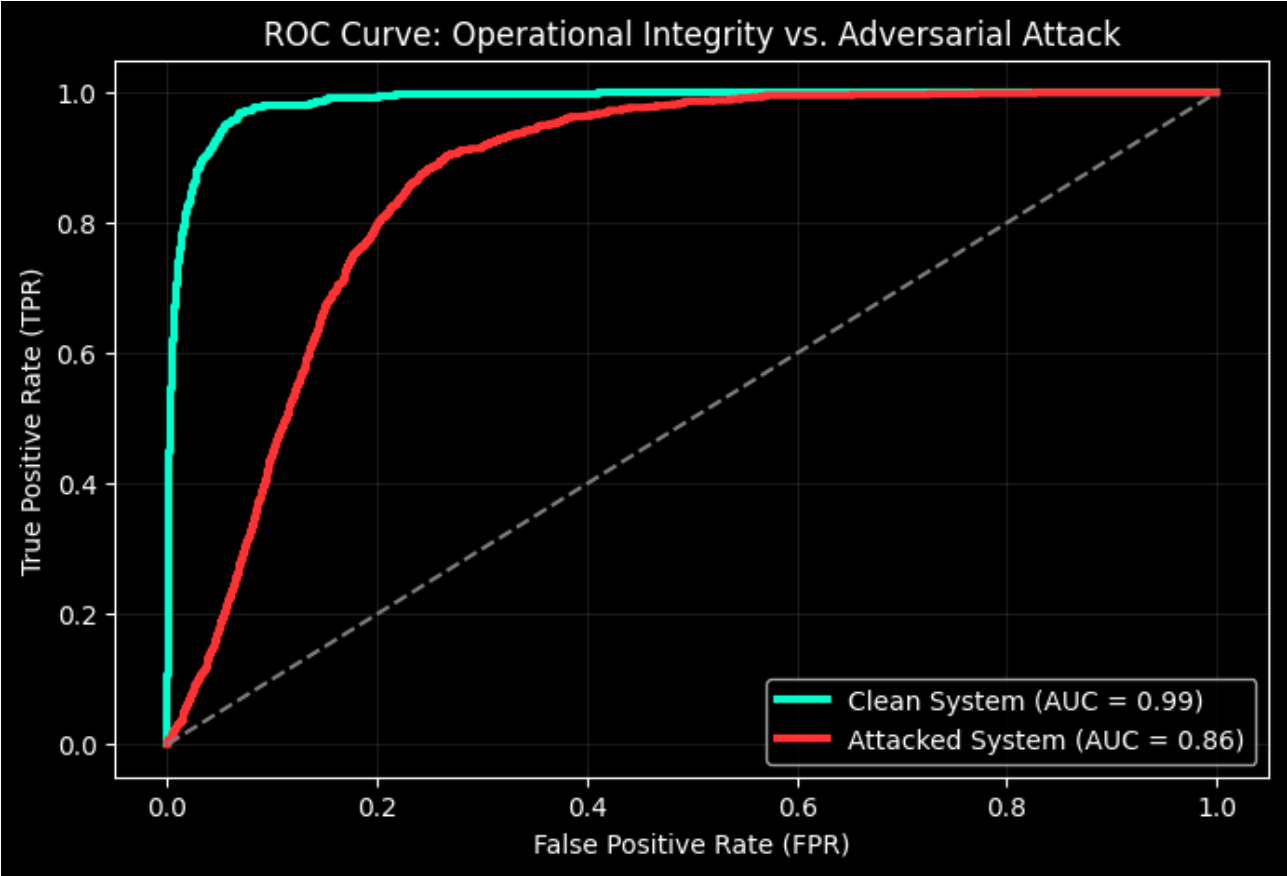
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SECTION 2: RELIABILITY & DISCRIMINATION ANALYSIS

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DESCRIPTION: The ROC Curve (Receiver Operating Characteristic) measures the system's total ability to distinguish between different structural patterns. The 'Area Under Curve' (AUC) represents the probability that the system will rank a correct feature higher than an incorrect one. A drop in AUC proves a fundamental loss of system integrity.



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SECTION 3: HARDWARE RELIABILITY AUDIT LOG (1,000 TEST SAMPLES)

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Metric	Clean System		Attacked System	
Accuracy	91.40	%	27.20	
Precision (1-FPR)	91.66	%	30.65	
Recall (TPR)	91.52	%	27.81	
F1-Score	91.54	%	28.21	

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[EXPLANATION OF METRICS]

- * ACCURACY: Overall system correctness.
- * PRECISION: Inverse of FPR. High precision means fewer False Alarms.

* RECALL (TPR): Sensitivity. Measures ability to catch ALL defects.
* F1-SCORE: The harmonic mean; best balance of Precision and Recall.
* ROC/AUC: Measures the system's discrimination power between classes.

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SECTION 4: SYSTEM HARDENING (THE SOLUTION)
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DESCRIPTION: Implementing Adversarial Training. We retrain the system with calculated noise to 'vaccinate' it against future FGSM attacks.

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

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    super().__init__(activity_regularizer=activity_regularizer,
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```

Hardening system... (Retraining with adversarial examples)

32/32 **0s** 3ms/step

32/32 **1s** 14ms/step

Vulnerable Accuracy under Attack: 27.20%

Hardened Accuracy under Attack: 99.30%

SECURITY IMPROVEMENT: 72.10%

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